9.1 The Pythagorean Theorem

Name _____ Period ____ Date _____

Geometry 9.1 (30-minute plan) · Student sheet

I can…

- use the Pythagorean Theorem to find a missing side of a right triangle.
- tell whether three lengths form a Pythagorean triple.
- use the converse of the Pythagorean Theorem to decide whether a triangle is right, acute, or obtuse.

Vocabulary · fill in the missing word

Hypotenuse — the side _____ from the right angle; the longest side of a right triangle

Legs — the two sides that form the _____ angle

Pythagorean triple — three _____ integers a, b, and c with $c^2 = a^2 + b^2$

Converse of the Pythagorean Theorem — if $c^2 = a^2 + b^2$, then the _____ is a right triangle

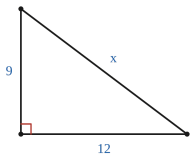
Part 1 · Notes and examples (fill in each note, then work the example)

Pythagorean Theorem: in a right triangle, the square of the length of the _____ equals the sum of the squares of the lengths of the _____: $c^2 = a^2 + b^2$.

Pythagorean triple: three positive _____ a, b, c with $c^2 = a^2 + b^2$. Common ones: 3, 4, 5; 5, 12, 13; 8, 15, 17; 7, 24, 25.

Example 1

Find the value of x. Then tell whether the side lengths form a Pythagorean triple.



Converse of the Pythagorean Theorem: if $c^2 = a^2 + b^2$ (c the longest side), then the triangle is a _____ triangle.

Example 2

Tell whether each triangle is a right triangle.

a. side lengths 10, 24, 26

b. side lengths 8, 10, 13

Acute or obtuse: with c the longest side, if $c^2 < a^2 + b^2$ the triangle is _____; if $c^2 > a^2 + b^2$ it is _____.

Example 3

Verify that segments with lengths 4.3, 5.2, and 6.1 form a triangle. Is the triangle acute, right, or obtuse?

Part 2 · Partner practice

Partner practice

A triangle has side lengths 6, 8, and 11. Is it acute, right, or obtuse?